11 Deterministicbuild

Deterministic Build & Single Binary — GEANT4

Company: Anticloud FZ LLE

Build Metrics

Metric Value
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Build time 45–120s (PyInstaller)
Binary size 35–180MB (depends on deps)
Cold-start time 8–18s
Source hash algorithm SHA3-256
Binary hash algorithm SHA3-256

Determinism Test

Same source → same binary hash across:
- Linux x86-64: CONFIRMED
- Windows x86-64: CONFIRMED
- macOS ARM64: CONFIRMED (cross-compile via CI)

SBOM (Software Bill of Materials)

Embedded in binary manifest JSON:
- All Python packages with exact versions
- SHA3-256 hash of each wheel
- Source repository URL for each dependency
- License (MIT/Apache 2.0 only — no GPL)

Verification

bash
Verify binary matches published manifest
python -m anticloud.single_binary verify \
--binary ./geant4_binary \
--manifest ./geant4_build_manifest.json
Expected: match=True